St. Joseph's College of Engineering and Technology, Palai

Department of Computer Applications

24SJMCA245 – Mini Project

QR based navigation app

Submitted By:	**Guided By:**
Student Name: Deepak Jose Thomas Reg No.: SJC25MCA - 2025 Roll No.: 25 Batch: 2025-2027	Name of Guide: Prof.Neethu Paulose Designation: Assistant Professor Department: MCA

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ABSTRACT

The **QR Code Based Classroom Navigation App** is a smart indoor navigation system designed to help students, faculty, and visitors locate classrooms easily within a college or university campus. The application works by scanning a QR code placed at different locations inside the campus. Once the QR code is scanned, the app identifies the user's current location and displays a digital map showing the shortest route to the selected classroom. It also provides step-by-step navigation with directional guidance, reducing confusion and saving time. The system is especially useful for new students, visitors, and during examinations when finding classrooms quickly is essential. The application offers a user-friendly interface, real-time navigation, and accurate location tracking within the campus.

Modules

1. User Authentication Module

- User registration and login.
- Secure authentication using email or student ID.
- Stores user profiles and preferences.

2. QR Code Scanning Module

- Scans QR codes placed at various campus locations.
- Detects the user's current position.
- Sends location data to the navigation system.

3. Classroom Search Module

- Allows users to search classrooms by room number, department, or faculty.
- Displays classroom details and availability.
- Provides destination selection.

4. Indoor Map Module

- Displays the building layout and floor maps.
- Highlights the user's current location.
- Supports zooming and interactive navigation.

5. Path Finding and Navigation Module

- Calculates the shortest and most efficient route.
- Uses pathfinding algorithms such as Dijkstra's or A*.
- Provides turn-by-turn directions (e.g., "Go straight," "Turn left," "Take the stairs").

6. Direction Guidance Module

- Gives visual directional indicators using arrows.
- Updates navigation based on the user's movement.
- Alerts users when they reach their destination.

7. Database Management Module

- Stores classroom information, QR code locations, maps, and user data.
- Maintains building layouts and navigation paths.
- Ensures quick retrieval of navigation information.

8. Admin Module

- Manages classroom details and building maps.
- Generates and updates QR codes.
- Adds, edits, or removes classrooms and routes.
- Monitors application usage and maintenance.

Technologies Used

Frontend

- Flutter (Dart) or React Native
- HTML, CSS, JavaScript (for web version)

Backend

- Node.js with Express.js / Django / Spring Boot

Database

- MySQL
- Firebase Firestore (optional for real-time updates)

QR Code Integration

- ZXing (Zebra Crossing)
- ML Kit Barcode Scanner
- Flutter QR Scanner Package

Maps and Navigation

- Google Maps SDK
- Indoor Map APIs
- A* or Dijkstra's Algorithm for shortest path calculation

Development Tools

- Android Studio
- Visual Studio Code
- Git & GitHub

Key Features

- QR code-based indoor navigation.
- Interactive digital campus map.
- Classroom search by room number or department.
- Shortest route calculation.
- Step-by-step directional guidance.
- Easy-to-use interface.
- Admin panel for updating maps and classroom information.
- Fast and accurate classroom location assistance.